

Higher-Order Differential Equations (Chapter 4)

Homogeneous Linear Equations with Constant Coefficients:

Homogeneous Linear Equation, $ay' + by = 0$; where the coefficients $a \neq 0$ and b are constants.

Higher-Order Linear Homogeneous Differential equation:

$$a_n y^n + a_{n-1} y^{n-1} + \cdots + a_1 y' + a_0 y = 0$$

where the coefficients $a_i, i = 0, 1, \dots, n$ are real constants and $a_n \neq 0$.

For example, let us consider 2nd order linear homogeneous differential equation,

$$ay'' + by' + cy = 0$$

Let the solution of the differential equation be $y = e^{mx}$ and we obtain the auxiliary equation.

Auxiliary equation: $am^2 + bm + c = 0$

After solving the auxiliary equation, we get $m_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $m_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

- ❖ If $(b^2 - 4ac > 0)$ then m_1 and m_2 are real and distinct.
- ❖ If $(b^2 - 4ac = 0)$ then m_1 and m_2 are real and equal.
- ❖ If $(b^2 - 4ac < 0)$ then m_1 and m_2 are conjugate complex numbers.

Distinct Real Roots: Solution, $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

Repeated Real Roots: Solution, $y = c_1 e^{mx} + c_2 x e^{mx}$

Conjugate Complex Roots: Solution, $y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$

$$\longrightarrow y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

Example 1:

Solve the following differential equations.

$$(a) \ 2y'' - 5y' - 3y = 0 \quad (b) \ y'' - 10y' + 25y = 0 \quad (c) \ y'' + 4y' + 7y = 0$$

Example 2:

Solve $4y'' + 4y' + 17y = 0, y(0) = -1, y'(0) = 2$.

Example 3:

$$\text{Solve } y''' + 3y'' - 4y = 0.$$

Example 4:

$$\text{Solve } \frac{d^4y}{dx^4} + 2 \frac{d^2y}{dx^2} + y = 0.$$

Exercise: Chapter 4.3: 3, 6, 9, 11, 13, 15, 31, 32, 37, 40

Undetermined Coefficients—Superposition Approach: To solve a nonhomogeneous linear differential equation,

$$a_n y^n + a_{n-1} y^{n-1} + \cdots + a_1 y' + a_0 y = g(x)$$

we must do two things:

- find the complementary function y_c and
- find any particular solution y_p of the nonhomogeneous equation

And the general solution is, $y = y_c + y_p$

Method of Undetermined Coefficient: The underlying idea behind this method is a conjecture about the form of y_p , an educated guess really, that is motivated by the kinds of functions that make up the input function $g(x)$. That is, $g(x)$ is a linear combination of functions of the type

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, p(x)e^{\alpha x}, p(x)e^{\alpha x} \sin \beta x \text{ and } p(x)e^{\alpha x} \cos \beta x$$

The method of undetermined coefficients is not applicable when

$$g(x) = \ln x, \quad g(x) = \frac{1}{x}, \quad g(x) = \tan x, \quad g(x) = \sin^{-1} x$$

Example 1:

$$\text{Solve } y'' + 4y' - 2y = 2x^2 - 3x + 6.$$

Example 2:

$$\text{Find a particular solution of } y'' - y' + y = 2 \sin 3x.$$

Example 3:

$$\text{Solve } y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}.$$

TABLE Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

Exercise: Chapter 4.4: 3, 5, 6, 7, 11, 14, 16, 18, 20, 24, 26

Variation of Parameters: Variation of parameters consists of finding a particular solution of the form $y_p = u_1(x)y_1(x)$. In other words, we have replaced the parameter c_1 by a function $u_1(x)$.

Summary of the Method: To solve the following equation

$$a_2y'' + a_1y' + a_0y = g(x)$$

- ❖ find the complementary function $y_c = c_1y_1 + c_2y_2$
- ❖ compute the Wronskian $W(y_1(x), y_2(x))$
- ❖ dividing by a_2 , we put the equation into the standard form $y'' + Py' + Qy = f(x)$ to determine $f(x)$.
- ❖ We find u_1 and u_2 by integrating $u'_1 = W_1/W$ and $u'_2 = W_2/W$
- ❖ A particular solution is $y_p = u_1(x)y_1(x)$.
- ❖ The general solution of the equation is then $y = y_c + y_p$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

Example 1:

Solve $y'' - 4y' + 4y = (x + 1)e^{2x}$.

Example 2:

Solve $4y'' + 36y = \csc 3x$.

Exercise: Chapter 4.6: 3, 6, 9, 10-13, 16, 21, 22, 27, 28